

# Workspace Efficiency Review

Tools, software and processes behind the Ryan PDS / EvAlx production environment

Prepared 17 September 2026 for Ryan PDS

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**Purpose.** Assess every layer of the current working environment (the Windows production PC, the Ableton MCP producer stack, the Hermes VPS, the AI agent fleet, storage and communication surfaces) and recommend concrete changes that make it faster, more reliable and better at producing finished tracks.

**Bottom line.** The environment is extremely capable but has grown sideways faster than it has been consolidated. Most lost time today comes from four sources: too many overlapping AI tools and chat surfaces, no enforced quality gate before a track is called finished, weak persistence (unsaved sets, almost no git history), and a nearly full system drive. Fixing those four delivers most of the gain.

## 1. Executive summary

Your stated goal for the whole system is to help you finish music: free-party acidcore, mental and tribe at 155 to 168 BPM, kick weight first, built in Ableton with Drift and the ESX-1. Measured against that goal, the environment has excellent raw ingredients. A capable PC (Ryzen 5 3600X, 32 GB RAM, RTX 4060), a working Ableton MCP bridge with a now-fixed Python client, a Tailscale link to a VPS, a Suno harvest pipeline that can turn stems and MIDI into arranged Ableton sets, and a documented agent fleet with clear lanes on paper.

The weaknesses are about focus and closure rather than capability. During the last 24 hours of work alone, the following happened: a finished-sounding track was posted to Slack with kick and bass clashing, because the only verification was RMS and peak statistics instead of your own mix ear; two instances of Ableton Live ran at once, which caused a long hunt for the right project; entire arrangements existed only in unsaved memory; and a 170-second file search timed out because nothing indexes the drive. None of these is a hard technical problem. Each one is a missing guardrail.

### Top ten recommendations

| #  | Recommendation                                                                                                                                                        | Impact    | Effort |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------|
| 1  | Make a mix QC gate mandatory before any track is called done or shared (kick/bass overlap, low-end mono, LUFS, true peak), using your ear.py / pocketscan.py mix ear. | Very high | Medium |
| 2  | Enforce a single Ableton instance and add save and export commands to the Remote Script so every build ends saved to a named file.                                    | Very high | Low    |
| 3  | Commit the ableton-mcp repo now (1 commit, 55 uncommitted changes) and adopt a daily commit habit with a secret scanner.                                              | High      | Low    |
| 4  | Free at least 100 GB on C: (currently 91 percent full, 40.7 GB free) and move audio projects off OneDrive sync.                                                       | High      | Low    |
| 5  | Cut the AI tool roster to one producer agent, one coding CLI and one local model runtime; uninstall or archive the rest.                                              | High      | Medium |
| 6  | Use one command surface and one notification channel instead of Slack, Telegram, Hermes WebUI, OpenCode and Cowork in parallel.                                       | High      | Medium |
| 7  | Wrap the harvest, render, build and verify scripts into one parameterised producer CLI command per stage.                                                             | High      | Medium |
| 8  | Install an instant file index (Everything by voidtools) and point agents at its CLI instead of recursive searches.                                                    | Medium    | Low    |
| 9  | Disable failing local MCP plugins (Prisma, CockroachDB toolbox) and keep one canonical copy of the Remote Script.                                                     | Medium    | Low    |
| 10 | Run all work through a small Kanban with a work-in-progress limit of one active track, matching the One Focus principle you already defined.                          | High      | Low    |

## 2. Scope and method

This review is based on direct inspection of the production PC (DESKTOP-4O4CU12) on 17 September 2026 and on the documented setup of your wider system. On the PC I measured running processes and memory, disk usage, the layout of the ableton-mcp repository and its git state, the .producer brain and tools, the Documents and Desktop folders, and the state of local MCP servers. I also drew on the concrete failures observed while building mental-acid-001 and EvAlx\_HarvestMix\_162 over the previous day, which are the best evidence of where the workflow actually breaks.

The Hermes VPS, Google Drive hubs and GitHub repositories were not inspected live in this pass, so recommendations about them rest on your documented architecture rather than fresh measurement. Your earlier audits (audit-2026-09-15.md, audit-2026-09-16.md) and testing-strategy-2026-09-17.md in the docs folder already cover bridge repair and test coverage; this report deliberately focuses on the wider workflow and avoids repeating them.

## 3. Current-state inventory

The table below summarises what is in place today, grouped by layer, with the key observation for each.

| Layer              | What is in place                                                                                                                                                       | Key observation                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Hardware           | Ryzen 5 3600X, 32 GB RAM, RTX 4060, single C: drive                                                                                                                    | C: has 435 GB used and only 40.7 GB free (about 91 percent full). Audio work at this scale needs headroom. |
| DAW                | Ableton Live 12 Suite, AbletonMCP Remote Script v1.7.0 on port 9877                                                                                                    | Two Live processes running (about 2.4 GB). No save or bounce command in the bridge.                        |
| Producer core      | ableton-mcp repo, MCP_Server/producer, .producer brain (3.1 GB of projects), abl.py client                                                                             | Only 1 git commit, 55 uncommitted changes, loose versioned scripts at repo root.                           |
| Analysis           | ai-music-ear on the VPS; ear.py, keyprobe.py, pocketscan.py mix ear; key_probe.py on PC                                                                                | Mix ear was not found in the PC tree and was not used in the build loop.                                   |
| AI agents and CLIs | Claude (10 processes), OpenCode (7), Codex, Cursor, Copilot, Gemini, Grok, Continue, Antigravity, Codely, OpenClaw, Pi, Ollama, llama.cpp, 9Router, cc-switch and more | Roughly twenty agent config folders in the home directory; heavy overlap in purpose.                       |
| Control and chat   | Slack (crisp, Hermes, ChatGPT, Claude, opencode bots), Telegram, Hermes WebUI, Cowork                                                                                  | Several parallel places to give commands and receive results.                                              |
| Server side        | Hermes VPS (hermex) with CRISP :8080, Music OS :5174, bridge :7843, LumaWave :5173, freellmapi :3001                                                                   | Some services belong to deprioritised projects but still run.                                              |
| Storage and sync   | OneDrive (Desktop redirected), Google Drive, Proton Drive, Tresorit, Splice                                                                                            | Four general sync services. Ableton projects landed in a OneDrive Untitled Project folder.                 |
| Local MCP plugins  | desktop-commander, gitkraken (working); Prisma, CockroachDB toolbox                                                                                                    | Two plugins fail on every start.                                                                           |

## 4. Findings and recommendations

### 4.1 Quality control: the missing gate before "done"

This is the most important finding because it directly affects the music. `EvAlx_HarvestMix_162` was verified by capturing the master output and checking RMS, peak and rough frequency bands. Those numbers said the track was loud and not silent. They could not say the kick and bass were fighting, and that is exactly what you heard. Your own mix ear (`ear.py` and `pocketscan.py`, which reuses `ear.py` band edges) exists to catch this class of problem, but it was not on the PC build path and nothing required it to run.

There is also a structural cause. The harvest approach layered a stem drum bus, a 909 lane, a KICK track and a KICKBASS track, plus a Drift acid bass and stem bass, without a rule for which element owns the sub region in each section. Kick weight first is your core taste principle, so it should be encoded as a hard rule rather than left to chance.

**Recommendation:** Define a QC gate that every render must pass before it is posted, shared or reported as finished. At minimum it should measure kick and bass energy overlap in the 30 to 120 Hz region on each kick hit, low-end phase and mono compatibility below 120 Hz, integrated LUFS and true peak, and a sub ownership check (only one element with sustained energy under 60 Hz at a time). Fail means the agent fixes and re-captures; it never posts.

**Recommendation:** Bring `ear.py`, `keyprobe.py` and `pocketscan.py` into the PC producer tree (for example `MCP_Server/producer/ear/`) and expose them as MCP tools such as `ear_scan(file)` and `ear_kick_bass(file)`, so every agent calls the same ear instead of improvising statistics.

**Recommendation:** Add a default low-end template to every generated set: sidechain compression or volume ducking on bass tracks keyed from the kick, a high-pass on all stem layers and pads, Utility Bass Mono on the bass bus, and a Limiter on the master. Because the API cannot reach the master track, save these in an Ableton default set (Live Set template) so they exist before the agent starts.

### 4.2 Ableton session hygiene and persistence

Several hours in the last day went to recovering from session state problems. Two Ableton Live processes were running; only one was bound to the MCP socket, so opening a file in the other made the title bar and the bridge disagree. Saves went to generic names (`tek.als`, `Untitled.als`, `Untitled2.als`) inside `OneDrive\Bureaublad\Untitled Project` because Windows redirects the Desktop to OneDrive. At the end of the session, a 200-bar mental-acid-001 build and a 192-bar HarvestMix build still existed only in memory.

**Recommendation:** Add a single-instance guard. A short PowerShell launcher that refuses to start Live if Ableton Live 12 Suite is already running, and a check in the producer CLI that aborts with a clear message when more than one Live process exists.

**Recommendation:** Extend the Remote Script with `save_set(path)` and, where Live allows it, an export or resample-to-audio command. Make every build script end with a save to a deterministic

name such as `EvAlx_<Concept>_<BPM>_<yyyymmdd-hhmm>.als`, then verify the file by decompressing it and checking tempo and track names, which is already proven to work.

**Recommendation:** Set a fixed Ableton project root outside OneDrive, for example `D:\Music\Projects` or `C:\Music\Projects` until a second drive exists, and set it as the default save location in Live preferences. Keep OneDrive for documents, not live audio projects, since sync locks and placeholder files interfere with large audio sessions.

**Recommendation:** Keep one canonical Remote Script. Today there is a patched copy in `Documents\Ableton\User Library` and an unpatched OneDrive mirror. Deploy from the repo (`AbletonMCP_Remote_Script`) with a script, and remove or ignore the mirror.

### 4.3 Tool sprawl: AI agents, CLIs and model runtimes

The home directory contains configuration for roughly twenty AI coding or agent tools, including Codex, Cursor, Copilot, Gemini, Grok, Continue, Antigravity, Codely (two variants), OpenClaw, Pi, NadirClaw, NoTokenLimit, OneProvider, cc-switch, Headroom, 9Remote, Ollama and `llama.cpp`, alongside Claude and OpenCode. At the time of measurement, Claude was running 10 processes using about 1.6 GB and OpenCode 7 processes using about 1.4 GB. Your own notes already concluded that Slack bots caused orchestration chaos and that the PC producer path should win when agents conflict; the same logic applies to the tool roster.

Every extra agent carries hidden costs: another place where instructions live, another config that drifts, another set of credentials, another process competing for RAM that Ableton needs, and another source of contradictory edits to the same files.

**Recommendation:** Consolidate to three roles. One primary producer agent that owns Ableton MCP and the producer core. One coding CLI for repository work. One local model runtime (choose Ollama or `llama.cpp`, not both) routed through 9Router, which already acts as your single model gateway. Everything else is archived: export any configs worth keeping into the repo docs, then uninstall.

**Recommendation:** Write the role map into one short file, for example `FLEET_ROLES.md` in the `ableton-mcp` docs, listing each surviving tool, what it may touch, and what it must never touch. Link it from `PRODUCER.md` so every agent reads the same boundaries.

**Recommendation:** Close agent sessions when idle. Seventeen background agent processes during an Ableton session is avoidable memory pressure. A scheduled task can report or end orphaned CLI processes older than a set age.

### 4.4 Communication and control surfaces

Commands and results currently flow through Slack (with crisp, Hermes, ChatGPT, Claude and opencode bots in the same channel), Telegram (with deliberate quiet rules and a single poller), Hermes WebUI (your main UI), OpenCode, and Cowork. A request like "send it to my Telegram bot" ending up in Slack, and uncertainty about whether a Slack post arrived, are symptoms of too many surfaces rather than bugs in any one of them.

**Recommendation:** Pick one command surface (where you tell the system what to do) and one notification surface (where finished, QC-passed results arrive). Given your notes, Hermes WebUI

or Cowork as the command surface and a single Slack channel for results is a natural split, with Telegram reserved for urgent alerts only.

**Recommendation:** Split #aimusicgeneration into a results channel that only receives QC-passed renders with a standard message template (title, BPM, key, LUFS, true peak, QC status, file link) and a separate bots or ops channel for logs, joins and experiments.

## 4.5 Version control and repository structure

The ableton-mcp repository is the documented source of truth for producer work, yet git shows a single commit ("hygiene: version bump") and 55 uncommitted changes. The repo root also mixes production code with one-off scripts: `continue_v9.py`, `continue_v10.py`, `rebuild_v9.py`, `events_v9.json`, `run_v9_variants.ps1`, `run_v10_variants.ps1`, `query_opencode_db.py` and `query_opencode_db2.py`, and `find_vps_pass.py` and `find_vps_pass2.py`. Scripts whose names suggest locating credentials should never sit in a repository root, even a private one.

**Recommendation:** Commit today, in logical chunks: the fixed `abl.py` client and smoke test, the Remote Script patch, the harvest-mix-162 pipeline, and docs. Add a pre-commit hook with `gitleaks` or `detect-secrets` so credential strings cannot be committed, which also enforces your existing redaction rule.

**Recommendation:** Restructure the root. Move reusable tools into `.producer/tools` or a `scripts/` folder, move versioned experiments into an `archive/` folder or delete them once committed, and move credential or database query helpers out of the repo entirely.

**Recommendation:** Decide what `.producer/projects` is. At 3.1 GB of rendered WAVs it should not be in git. Keep the scripts and manifests in git, ignore `render/`, `source/` and `verify/` audio, and back those up with a sync tool instead.

## 4.6 Storage, files and disk capacity

The system drive is the single biggest performance and reliability risk. With 40.7 GB free on a 476 GB drive, one more stems pack plus renders can push it into the zone where Windows, OneDrive and Ableton all slow down or fail writes. Clear duplication exists today: the Suno harvest archive is stored both as 32 split parts (1.3 GB) and as the joined zip (1.3 GB), and the Gabber time stems zip (352 MB) sits beside its extracted copy. Downloads holds 229 items, and the OneDrive Desktop holds executables, zipped workspaces, master WAVs and screenshots.

Storage is also spread across OneDrive, Google Drive, Proton Drive, Tresorit and Splice. Each has a legitimate use, but without a rule for which data goes where, files are found by searching rather than by knowing.

**Recommendation:** Reclaim space now. Delete the 32 zip parts and the join script once the joined zip is verified, delete extracted-and-harvested zips, and run `WizTree` to find the largest remaining folders (Ableton caches, uv and npm caches, old model files under `.ollama` are common culprits).

**Recommendation:** Add a dedicated audio drive (a 1 to 2 TB NVMe or SATA SSD) for Ableton projects, sample libraries, stems and renders. This is the highest value hardware purchase for this workflow.

**Recommendation:** Adopt a storage map and write it down: OneDrive for documents only, Google Drive for shared fleet hubs, Tresorit or Proton for secrets and personal files, the audio drive for all music, Splice for samples. Consider dropping one of Proton Drive or Tresorit, since they overlap.

**Recommendation:** Use a fixed inbox pattern: new packs land in Music\Inbox, get harvested, then move to Music\Library\<pack> or get deleted. Never harvest directly from Documents.

## 4.7 Reliability of the automation layer

Several small reliability issues each cost minutes and together cost hours. Recursive searches through the home directory timed out at 170 seconds. The device bridge dropped twice mid-build. The Prisma and CockroachDB toolbox MCP plugins fail every time they start. The Remote Script had a missing method and a note-timing key mismatch that silently placed every note at beat zero until it was found. `create_locator` fails when the transport is stopped.

**Recommendation:** Install Everything (voidtools) and its `es.exe` command line. Agents can then locate any file on the PC in milliseconds instead of walking the drive.

**Recommendation:** Disable or uninstall the Prisma and CockroachDB plugins unless a current project uses them. Failing servers add startup delay and log noise that hides real errors.

**Recommendation:** Turn the smoke test into a pre-flight check that every build runs first: socket reachable, exactly one Live instance, tempo set, note round-trip with `start_time` verified, browser URIs resolvable, free disk above a threshold. Build scripts refuse to start if pre-flight fails.

**Recommendation:** Make build scripts resumable. Write a small state file per project after each phase (tracks created, clips placed, trims applied), so a dropped connection resumes from the last completed phase instead of re-running or leaving a half-applied rebalance.

## 4.8 Repetitive work that should be automated

The last day repeated the same sequence several times by hand or with one-off scripts: extract a pack, analyse tempo maps and keys, find repeating hooks, render tempo-locked audio, build tracks in Ableton, capture loopback audio, analyse it, trim levels, then post to Slack. The pieces are good and already proven (`tempomap.py`, the Jaccard hook finder, per-bar rubberband rendering, loopback capture). They are just not packaged.

| Repeated task            | Current state                                                | Automation                                                     |
|--------------------------|--------------------------------------------------------------|----------------------------------------------------------------|
| Pack intake and analysis | Per-project scripts 01 and 02                                | producer harvest <pack> writes analysis.json and manifest.json |
| Arrangement build        | Per-project 03_build_ableton.py with hard-coded section maps | producer build <manifest> --map <sections.yaml>                |
| Verification capture     | Ad-hoc loopback scripts                                      | producer capture --bars 1,49,97 feeding the mix ear            |
| Mix QC                   | Not automated                                                | producer qc <capture> returns pass or fail with reasons        |
| Save and backup          | Manual, often forgotten                                      | Automatic save at end of build plus nightly copy of projects   |
| Sharing                  | Manual Slack upload flow                                     | producer publish runs only after QC pass,                      |

| Repeated task            | Current state | Automation                                                                      |
|--------------------------|---------------|---------------------------------------------------------------------------------|
|                          |               | using the standard template                                                     |
| Disk and process hygiene | Manual        | Weekly scheduled task: free space report, orphan agent processes, cache cleanup |

**Recommendation:** Build the producer CLI around those stages, reusing MCP\_Server/producer/cli.py, which already exists. Store section maps as YAML so a new arrangement is a data change, not a code change.

## 4.9 Focus, planning and knowledge management

You have defined good principles: finish music rather than generate it, One Focus, Kanban, and memory features designed around ADHD. In practice, in one day work jumped between mental-acid-001, gabber-time-remake (later abandoned), a new HarvestMix, a Slack delivery, and a workspace audit. Several projects are marked deprioritised (Music OS, LumaWave, Vestiges sonic lane) but their services still run on the VPS. There is also a direct conflict between the taste rule "do not build from gabber, hardstyle or trance listen-lanes" and a new project built from a gabber stem pack, which is fine as a conscious exception but should be recorded as one.

**Recommendation:** Keep a single board (Notion is already connected, or a plain Kanban file in the repo) with columns Inbox, Next, Now, QC, Done, and a hard limit of one track in Now. Agents may only work on the card in Now unless you move another card.

**Recommendation:** Keep one decision log per project with dated one-line decisions, for example "2026-09-17: gabber stems used as texture only, exception to taste rule". This prevents agents from re-arguing settled choices or silently breaking rules.

**Recommendation:** Shut down or park services for deprioritised projects on the VPS (Music OS :5174, LumaWave :5173, bridge :7843 if unused) and note in the board how to restart them. Fewer running things means fewer things to monitor and secure.

**Recommendation:** Standardise documentation language. Audits are in Dutch, testing strategy and PRODUCER.md in English. Choose one for agent-facing docs so every model reads them consistently.

## 5. Recommended tools and methods

Each tool below addresses a specific finding above. Where a free option works well, it is listed first.

| Need                     | Tool or method                                                                  | Why it fits                                                  |
|--------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------|
| Loudness and true peak   | Youlean Loudness Meter (free plugin); pyloudnorm in the QC script               | Visible LUFS in Ableton plus the same numbers in automation. |
| Kick and bass separation | Ableton Compressor sidechain or a volume-shaper such as Kickstart 2 or LFO Tool | Enforces kick ownership of the sub region in every section.  |
| Low-end checks           | Ableton Utility (Bass Mono), EQ Eight spectrum, your ear.py band edges          | Already owned; needs to be templated and scripted.           |
| Set templates            | Ableton default Live Set at 162 BPM with                                        | Covers what the API cannot reach,                            |



| Need           | Tool or method                                                            | Why it fits                                                  |
|----------------|---------------------------------------------------------------------------|--------------------------------------------------------------|
|                | master Limiter and bus routing                                            | before the agent starts.                                     |
| File search    | Everything (voidtools) with es.exe                                        | Instant indexed search for humans and agents.                |
| Disk analysis  | WizTree                                                                   | Finds large folders in seconds on NTFS.                      |
| Secrets safety | gitleaks or detect-secrets as a pre-commit hook                           | Blocks credential commits automatically.                     |
| Backups        | rclone or Tresorit scheduled sync of the projects folder; git for scripts | Separates code history from audio backup.                    |
| Scheduling     | Windows Task Scheduler (already your pattern for detached work)           | Weekly hygiene and nightly backups without SSH sessions.     |
| Planning       | Notion board or a KANBAN.md with WIP limit 1                              | Makes One Focus a rule agents can read.                      |
| Hardware       | Second SSD, 1 to 2 TB, for audio                                          | Removes the disk-full risk and speeds sample-heavy sessions. |

## 6. Prioritised roadmap

### Today (under two hours)

- Save both open Ableton arrangements under proper names outside OneDrive and close the second Live instance.
- Commit the ableton-mcp repository and move credential-related scripts out of it.
- Delete the 32 Suno zip parts and other duplicate archives (about 1.7 GB), then run WizTree to find the bigger wins toward the 100 GB target.
- Disable the failing Prisma and CockroachDB plugins.

### This week

- Bring the mix ear into the PC producer tree and expose it as MCP tools; define QC thresholds for kick and bass overlap, LUFS and true peak.
- Fix the kick and bass clash in EvAlx\_HarvestMix\_162 using sub ownership per section and sidechain on bass tracks, then re-run QC.
- Create the 162 BPM Ableton default set with master Limiter and low-end template.
- Install Everything and WizTree; add a single-instance Ableton launcher.
- Write FLEET\_ROLES.md and uninstall the agent tools that fall outside the three roles.

### This month

- Package harvest, build, capture, QC and publish into the producer CLI with YAML section maps and resumable state.
- Add save\_set to the Remote Script and make every build end with a verified save.
- Buy and set up the audio SSD; migrate projects, packs and renders; write the storage map.

- Split Slack into results and ops channels; set the board with a WIP limit of one.
- Park deprioritised VPS services and record restart steps.

## 7. How to know it is working

Track a few simple numbers for four weeks. They show whether the changes are producing more finished music rather than more activity.

| Metric                                             | Today                                | Target             |
|----------------------------------------------------|--------------------------------------|--------------------|
| Tracks reaching QC pass per week                   | Not measured; QC gate does not exist | At least 1         |
| Renders shared that later fail your listening test | At least 1 in the last day           | Zero               |
| Builds ending with a verified saved .als           | About 1 of 3 in the last day         | Every build        |
| Free space on C:                                   | 40.7 GB                              | Above 120 GB       |
| AI agent tools installed                           | About 20                             | 3 roles            |
| Git commits in producer repo per week              | 1 total                              | Daily when working |
| Time from new pack to first playable arrangement   | Several hours with manual steps      | Under 30 minutes   |

## Appendix: evidence captured on 17 September 2026

- **Processes:** Ableton Live 12 Suite x2 (2,465 MB), claude x10 (1,568 MB), OpenCode x7 (1,394 MB), node x8 (750 MB), tailscaled x2, python x2, OneDrive, wslservice, sshd.
- **Disk C:** 435.2 GB used, 40.7 GB free.
- **Hardware:** AMD Ryzen 5 3600X, 31.9 GB RAM, NVIDIA GeForce RTX 4060.
- **Repository:** ableton-mcp has 1 commit and 55 uncommitted changes; root contains versioned experiment scripts and credential/database query helpers.
- **Producer data:** .producer/projects is 3.1 GB (acid-tekno-170, gabber-time-remake, harvest-mix-162, how-about-everyone-else, mental-acid-001).
- **Documents:** Suno\_Harvest\_ALL\_for\_PC.zip (1.3 GB) plus 32 part files (1.3 GB), Gabber time (Remastered) Stems.zip (352 MB), Acid Pulse MIDI and zip.
- **Local MCP:** desktop-commander and gitkraken announced; Prisma-Local and cockroachdb-toolbox failed.
- **Session incidents:** two concurrent Live instances; saves to OneDrive Untitled Project with generic names; unsaved 200-bar and 192-bar arrangements; kick and bass clash in a shared render; note start\_time key bug; missing get\_browser\_categories handler; device bridge drops; 170-second search timeout.